Haemopoiesis

HAEMOPOIESIS (sample notes)

All blood cells derive from haematopoietic stem cells (HSCs) in the bone marrow. HSCs are self-renewing and pluripotent.

Two main lineages
- Common myeloid progenitor: erythrocytes, megakaryocytes (platelets), granulocytes (neutrophils, eosinophils, basophils), monocytes (which become macrophages and dendritic cells).
- Common lymphoid progenitor: B cells, T cells, natural killer cells.

Key growth factors
- Erythropoietin (EPO), made by the kidney, drives red cell production.
- Thrombopoietin (TPO), mainly from the liver, drives platelet production.
- G-CSF stimulates neutrophil production.

Clinical link: anaemia of chronic kidney disease is often treated with erythropoiesis-stimulating agents because EPO production falls as kidney function declines.